

Experimental Methods in Solid State Physics

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1 Probing lattice vibrations with light

Lattice vibrations are quantized into phonons, with each vibrational mode (momentum $\mathbf{q}$, index or branch s) having energy $\hbar\omega_{\mathbf{q}s}$ and an occupancy $n = 0, 1, 2, \dots$. This quantization process assumes that the lattice vibrations can be treated as sufficiently long-lived **quasi-particle** excitations. Spectroscopic techniques that exchange small amounts of energy and momentum with the lattice provide experimental access to these quasi-particles. In this chapter we introduce two light-scattering techniques:

- **Raman scattering** (optical phonons, THz frequencies),
- **Brillouin light scattering** (acoustic phonons, GHz frequencies).

Both processes involve the creation or annihilation of phonons via inelastic scattering of photons. The measurable frequency shift of the scattered light directly reflects phonon energies.

Stokes and anti-Stokes processes. When an incident photon of angular frequency ω_{in} interacts with a lattice vibration, two processes are possible:

- **Stokes (phonon creation):**

$$\omega_{\text{out}} = \omega_{\text{in}} - \omega_{\mathbf{q}s},$$

the scattered photon loses energy and a phonon is created. This process is always allowed because the number of phonons is not a conserved quantity.

- **Anti-Stokes (phonon annihilation):**

$$\omega_{\text{out}} = \omega_{\text{in}} + \omega_{\mathbf{q}s},$$

the scattered photon gains energy by annihilating (or absorbing) an existing phonon. This process is only allowed if phonon are already thermally excited. Therefore, the anti-Stokes intensity is proportional to the thermal phonon occupation, $I_{\text{anti-Stokes}} \propto n(\omega_{\mathbf{q}s})$, which makes it a direct thermometer for vibrational populations.

Because visible photons carry only very small momentum $|\mathbf{k}| \sim 10^{-2} \text{ \AA}^{-1}$, optical scattering experiments typically probe phonons close to the Brillouin-zone centre ($\mathbf{q} \approx 0$). Raman scattering accesses predominantly *optical* phonons, while Brillouin scattering accesses predominantly *acoustic* phonons at small $\mathbf{q}$.

Normal modes, phonon quantization, and quasiparticles

Harmonic approximation. For small lattice vibrations, the potential energy of the crystal can be expanded around equilibrium positions. Keeping only quadratic terms in the displacements $\{\mathbf{u}_\ell\}$ gives the *harmonic Hamiltonian*

$$\hat{H}_2 = \sum_{\ell} \frac{\hat{\mathbf{p}}_{\ell}^2}{2M_{\ell}} + \frac{1}{2} \sum_{\ell, \ell'} \hat{\mathbf{u}}_{\ell}^T \mathbf{D}_{\ell\ell'} \hat{\mathbf{u}}_{\ell'},$$

where $\mathbf{D}_{\ell\ell'}$ is the *dynamical matrix*. This is a system of coupled harmonic oscillators.

Normal modes. Translational symmetry allows us to transform to collective coordinates labeled by a wave vector $\mathbf{q}$ and a branch index s . Diagonalizing the dynamical matrix yields *normal modes* with frequencies $\omega_{\mathbf{q}s}$:

$$\hat{H}_2 = \sum_{\mathbf{q}, s} \left[\frac{\hat{P}_{\mathbf{q}s}^2}{2} + \frac{1}{2} \omega_{\mathbf{q}s}^2 \hat{Q}_{\mathbf{q}s}^2 \right].$$

Each mode $(\mathbf{q}, s)$ behaves as an independent harmonic oscillator.

Quantization and phonons. Quantizing each normal mode introduces creation and annihilation operators $a_{\mathbf{q}s}^{\dagger}$ and $a_{\mathbf{q}s}$, yielding

$$\hat{H}_2 = \sum_{\mathbf{q}, s} \hbar \omega_{\mathbf{q}s} \left(a_{\mathbf{q}s}^{\dagger} a_{\mathbf{q}s} + \frac{1}{2} \right).$$

The state $a_{\mathbf{q}s}^{\dagger}|0\rangle$ is a *phonon*: a quantized lattice vibration with energy $\hbar\omega_{\mathbf{q}s}$ and crystal momentum $\hbar\mathbf{q}$. In the harmonic approximation, these excitations are *non-interacting* and have infinite lifetime.

Anharmonicity and lifetime. Real crystals include higher-order terms $\hat{H}_3, \hat{H}_4, \dots$ in the Hamiltonian. These terms allow phonons to scatter from each other (and from defects), broadening their spectral peaks. The phonon lifetime is

$$\tau_{\mathbf{q}s} = \frac{1}{\Gamma_{\mathbf{q}s}},$$

where $\Gamma_{\mathbf{q}s}$ is the interaction-induced linewidth. The quasiparticle description remains valid as long as $\Gamma_{\mathbf{q}s} \ll \omega_{\mathbf{q}s}$, ensuring a well-defined, long-lived mode.

Quasiparticle viewpoint. A phonon is therefore a *quasiparticle*: an emergent, weakly interacting excitation obtained by quantizing the normal modes of the harmonic crystal and then allowing small anharmonic corrections that give it a finite lifetime. Its properties include energy $\hbar\omega_{\mathbf{q}s}$, polarization, branch index s , crystal momentum $\hbar\mathbf{q}$, and lifetime $\tau_{\mathbf{q}s}$.

Crystal momentum vs. physical momentum. The quantity $\hbar\mathbf{q}$ is a *crystal momentum*, defined modulo any reciprocal lattice vector $\mathbf{G}$. It reflects the periodicity of the vibrational mode, not the actual mechanical momentum of atoms. Crystal momentum is conserved in processes that respect lattice periodicity and governs selection rules in light-matter and phonon-phonon interactions.

1.1 Inelastic light scattering and the dynamical structure factor

In an inelastic scattering event, an incoming photon with $(\mathbf{k}, \omega)$ is scattered into $(\mathbf{k}', \omega')$. The lattice absorbs or emits a phonon with

$$\hbar\mathbf{q} = \hbar(\mathbf{k} - \mathbf{k}'), \quad \hbar\omega_{\mathbf{q}s} = \hbar(\omega - \omega').$$

Phonon creation corresponds to $\omega' < \omega$ (Stokes), and phonon annihilation corresponds to $\omega' > \omega$ (anti-Stokes). The intensity of inelastically scattered light is proportional to the *dynamical structure factor* $S(\mathbf{q}, \omega)$, which for phonons contains distinct contributions from phonon creation and annihilation:

$$S(\mathbf{q}, \omega) \propto \sum_s |\mathbf{e}_{\mathbf{q}s} \cdot \hat{\mathbf{e}}|^2 [n(\omega_{\mathbf{q}s}) \delta(\omega + \omega_{\mathbf{q}s}) + (n(\omega_{\mathbf{q}s}) + 1) \delta(\omega - \omega_{\mathbf{q}s})].$$

- $\mathbf{e}_{\mathbf{q}s}$ is the polarization vector of the phonon mode (LA phonons $\mathbf{e}_{\mathbf{q}s} \parallel \mathbf{q}$, TA phonons $\mathbf{e}_{\mathbf{q}s} \perp \mathbf{q}$, optical phonons $\mathbf{e}_{\mathbf{q}s}$ describes out-of-phase motion between atoms in the same unit cell)
- $\hat{\mathbf{e}}$ is the unit polarization vector of the incident (or scattered) light.
- The term with $\delta(\omega - \omega_{\mathbf{q}s})$ describes **Stokes scattering**.
- The term with $\delta(\omega + \omega_{\mathbf{q}s})$ describes **anti-Stokes scattering**.
- $n(\omega_{\mathbf{q}s})$ is the Bose–Einstein phonon occupation:

$$n(\omega) = \frac{1}{e^{\hbar\omega/(k_B T)} - 1}.$$

Thus the ratio of anti-Stokes to Stokes intensities becomes

$$\frac{I_{\text{anti-Stokes}}}{I_{\text{Stokes}}} = e^{-\hbar\omega_{\mathbf{q}s}/(k_B T)},$$

providing a direct measure of thermal phonon populations. Raman and Brillouin scattering therefore probe not only phonon *energies* but also phonon *statistics*.

Bose-Einstein-statistics: occupation number of phonons

A single lattice vibration mode of frequency ω can carry an integer number $n = 0, 1, 2, \dots$ of phonons. Each phonon has energy $\varepsilon = \hbar\omega$, so that the total energy of the mode is

$$E_n = n \hbar\omega.$$

(For clarity we omit the constant zero-point energy $\frac{1}{2}\hbar\omega$, which cancels from all probability ratios.)

Thermal equilibrium. In the canonical ensemble, the probability of occupying a state of energy E_n is proportional to the Boltzmann factor,

$$P_n \propto e^{-E_n/(k_B T)} = e^{-n\hbar\omega/(k_B T)}.$$

Introduce the shorthand

$$x := e^{-\hbar\omega/(k_B T)},$$

The properly normalized probability distribution is therefore

$$P_n = \frac{x^n}{\sum_{m=0}^{\infty} x^m},$$

The normalization factor is the geometric series

$$Z = \sum_{m=0}^{\infty} x^m = \frac{1}{1-x},$$

which yields

$$P_n = (1-x) x^n.$$

Average phonon occupation. The thermal average of the phonon number is

$$\langle n \rangle = \sum_{n=0}^{\infty} n P_n = (1-x) \sum_{n=0}^{\infty} n x^n.$$

Using the identity

$$\sum_{n=0}^{\infty} n x^n = \frac{x}{(1-x)^2},$$

we obtain

$$\langle n \rangle = \frac{x}{1-x} = \frac{1}{e^{\hbar\omega/(k_B T)} - 1}.$$

Interpretation. The result

$$\langle n \rangle = \frac{1}{e^{\hbar\omega/(k_B T)} - 1}$$

is the Bose-Einstein distribution for a single phonon mode. It follows only from:

- energy quantization in steps of $\hbar\omega$,
- the Boltzmann weight $e^{-E_n/(k_B T)}$,
- and the fact that any integer number of phonons is allowed in the same state with energy (bosons have no exclusion principle).

1.2 Raman scattering

Raman scattering is an inelastic light-scattering process in which an incident photon excites a virtual electronic state and exchanges energy with a vibrational mode of the crystal. Because visible photons carry very small momentum, (first-order) Raman scattering probes phonons close to $\mathbf{q} \approx 0$, i.e. optical phonons at the Brillouin-zone centre.

Figure 1 shows the calculated phonon dispersion in silicon (Fig. 1a) along with an experimentally measured Raman spectrum (Fig. 1b). The energies Raman spectroscopy are referenced to the laser line. The typical unit used is inverse wavenumber cm^{-1} , whereby 1cm^{-1} corresponds to about 0.0297THz . The main Raman mode of silicon arises from the TO phonon at the Γ -point near 520cm^{-1} , but several second (and higher) order features are visible as well. These higher order processes can include contributions from finite momenta and acoustic phonons as well.

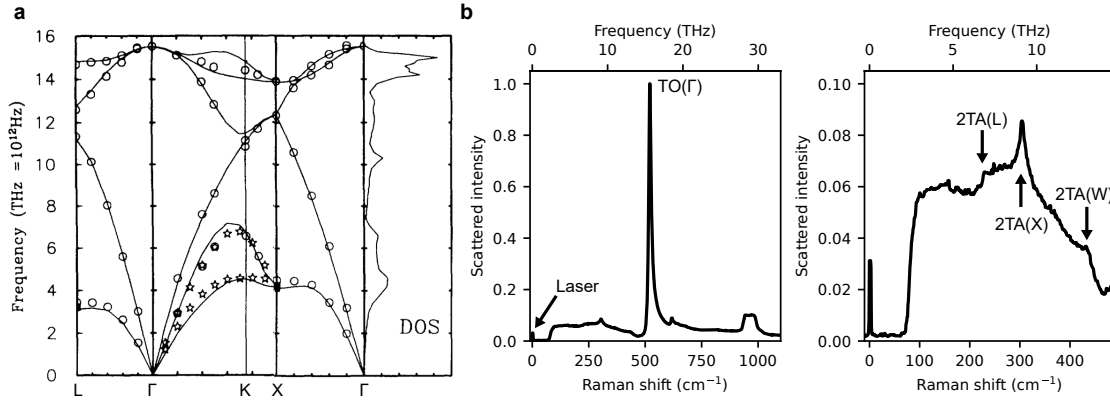


Figure 1: Phonons and Raman spectrum of silicon. **a** Phonon dispersion of Si along high symmetry directions. **b** Stokes part of the Raman spectrum of Si measured at an excitation wavelength of 532 nm (green laser).

Experimental implementation

Modern Raman experiments are typically performed in a microscope geometry. A narrow-band, continuous-wave laser (commonly 532 nm or 633 nm) is focused onto the sample through a high-numerical-aperture objective. The same objective collects the back-scattered light. A set of dichroic mirrors and notch or edge filters suppresses the intense elastic (Rayleigh) line by more than six orders of magnitude, allowing the much weaker Stokes and anti-Stokes signals to be detected. The frequency analysis is carried out using a grating spectrometer with a CCD detector. Raman microscopy combines optical spectroscopy with micron-scale spatial resolution, and it is widely applicable to semiconductors, oxides, organics, and 2D materials.

Origin of the Raman process

Raman scattering is a *two-step* light-matter interaction:

1. the incident photon excites the system into an intermediate (virtual) state;
2. during relaxation into the outgoing photon state, a phonon is created or annihilated.

The intermediate state does not need to coincide with a real electronic energy level; it can be reached only transiently due to the time–energy uncertainty relation. This is why Raman scattering occurs even when the photon energy is far from any real optical transition. The overall strength of a Raman mode is controlled by how strongly the polarizability of the material changes when the atoms are displaced along the phonon eigenvector. Only phonons that modulate the polarizability tensor are Raman active (Raman selection rules).

Light–matter interaction in the dipole approximation

Minimal coupling. An electron in an electromagnetic field is described by the Hamiltonian

$$\hat{H} = \frac{1}{2m} (\hat{\mathbf{p}} + e\mathbf{A}(\mathbf{r}, t))^2 + V(\mathbf{r}),$$

written here in Coulomb gauge ($\nabla \cdot \mathbf{A} = 0$, $\phi = 0$). The interaction with light originates from the cross term

$$\hat{H}_{\text{int}} = \frac{e}{2m} (\hat{\mathbf{p}} \cdot \mathbf{A}(\mathbf{r}, t) + \mathbf{A}(\mathbf{r}, t) \cdot \hat{\mathbf{p}})$$

plus a usually negligible A^2 term.

Dipole approximation. For optical wavelengths ($\lambda \sim 500$ nm) the spatial variation of the vector potential is tiny on the scale of atomic wave functions ($r \sim 0.1$ nm). Thus the electromagnetic field is effectively uniform across the unit cell:

$$\mathbf{A}(\mathbf{r}, t) \approx \mathbf{A}(t).$$

Under this assumption, $\mathbf{A}(t)$ commutes with $\hat{\mathbf{p}}$, and the interaction term simplifies to

$$\hat{H}_{\text{int}}(t) = \frac{e}{m} \hat{\mathbf{p}} \cdot \mathbf{A}(t).$$

This form is often called the *velocity gauge*.

Transformation to the dipole (length) gauge. A time-dependent gauge transformation (see quantum mechanics textbooks) can be used to eliminate the vector potential from the kinetic term when $\mathbf{A}$ is spatially uniform. The transformed Hamiltonian becomes

$$\hat{H} = \hat{H}_0 - \hat{\mathbf{d}} \cdot \mathbf{E}(t), \quad \hat{\mathbf{d}} = -e \hat{\mathbf{r}},$$

with $\mathbf{E}(t) = -\partial \mathbf{A} / \partial t$. This is the *dipole Hamiltonian*. It contains the same physics as the minimal-coupling form but makes the interaction with light explicit: the electric field couples directly to the dipole moment of the electrons.

Connection to polarizability. The expectation value of the dipole operator defines the macroscopic polarization,

$$\mathbf{P}(t) = \langle \hat{\mathbf{d}}(t) \rangle,$$

which for weak fields responds linearly to $\mathbf{E}(t)$:

$$P_i(\omega) = \sum_j \alpha_{ij}(\omega) E_j(\omega).$$

The tensor $\alpha(\omega)$ is the *polarizability*, whose derivatives with respect to vibrational coordinates generate inelastic light-scattering processes. In particular, a phonon-induced modulation of α produces Raman and Brillouin sidebands in the scattered light.

Rayleigh and Raman scattering

First-order processes: absorption, emission, Rayleigh scattering. In time-dependent perturbation theory with $\hat{H}_{\text{int}}(t)$, the transition matrix element between initial and final states $|i\rangle$ and $|f\rangle$ is, to first order,

$$M_{fi}^{(1)} \propto \langle f | \hat{\mathbf{d}} \cdot \hat{\mathbf{e}} | i \rangle,$$

with $\hat{\mathbf{e}}$ the light polarisation. This describes *single-photon* processes such as absorption, stimulated emission and elastic (Rayleigh) scattering, where the system goes directly from $|i\rangle$ to $|f\rangle$ in one step.

Raman scattering as a second-order process. Raman scattering is a *second-order* light-matter process: the system passes through an intermediate (virtual) state $|v\rangle$ that is not necessarily an eigenstate of the unperturbed Hamiltonian with the correct energy. In second-order time-dependent perturbation theory the transition amplitude is

$$M_{fi}^{(2)} \propto \sum_v \frac{\langle f | \hat{H}_{\text{int}} | v \rangle \langle v | \hat{H}_{\text{int}} | i \rangle}{\omega_{\text{in}} - (E_i - E_v)},$$

where E_i and E_v are the energies of $|i\rangle$ and $|v\rangle$. The intermediate state is “virtual” in the sense that it is only populated transiently and does not satisfy energy conservation by itself. Because Raman scattering involves two powers of the light-matter interaction, and because the intermediate state is usually far off resonance (large $|E_i - E_v|$ in the denominator), the resulting cross section is much smaller than for first-order dipole processes. Resonant Raman occurs if the incoming photon energy ω_{in} matches resonantly the energy of a real intermediate state ($E_i - E_v$).

Resonant vs. non-resonant Raman scattering

If the laser energy happens to lie close to a real electronic transition in the material, the intermediate state becomes quasi-resonant. In this case the Raman process is strongly enhanced:

- **Non-resonant Raman:** the intermediate state is far from real electronic transitions. The scattering probability is relatively weak and depends mainly on vibrational symmetry.
- **Resonant Raman:** the laser energy matches or approaches a real electronic excitation. The scattering amplitude increases dramatically, enhancing selected phonon modes.

A detailed description of electronic structure is not required here. The important point is that real electronic excitations act as “amplifiers” for the vibrational scattering process. Resonance can be tuned by choosing the laser wavelength, which is used, for example, to selectively enhance specific phonons in 2D materials or molecular crystals.

Two-phonon Raman scattering and zone-edge acoustic modes

Although first-order Raman scattering accesses only phonons with $\mathbf{q} \approx 0$, higher-order processes allow phonons from across the Brillouin zone to contribute. In two-phonon Raman scattering the photon creates (or annihilates) *two* phonons with momenta $\mathbf{q}_1$ and $\mathbf{q}_2$ such that

$$\mathbf{q}_1 + \mathbf{q}_2 \approx 0,$$

which satisfies momentum conservation for photons with very small momentum. The observed Raman shift is then

$$\omega = \omega_{\mathbf{q}_1 s_1} + \omega_{\mathbf{q}_2 s_2}.$$

This process enables access to *zone-edge acoustic phonons*. Although acoustic modes have $\omega=0$ at $\mathbf{q}=0$, they reach large frequencies near the Brillouin-zone boundary and can therefore appear in two-phonon Raman spectra. In silicon, for example, a broad band around $\sim 950 \text{ cm}^{-1}$ arises from combinations of transverse and longitudinal acoustic phonons near the X and L points. In layered materials such as graphene, double-resonant Raman scattering involving zone-edge acoustic modes gives rise to the prominent “2D” band. Two-phonon Raman scattering thus complements first-order scattering: it provides indirect information about phonons away from $\mathbf{q}=0$, including the upper parts of the acoustic branches.

Defect-activated Raman scattering

In an ideal crystal, first-order Raman scattering obeys the momentum conservation rule

$$\mathbf{q} \approx \mathbf{k} - \mathbf{k}' \approx 0,$$

so only zone-centre phonons contribute. Real crystals, however, contain point defects, dislocations, grain boundaries, or surface steps. Such imperfections break translational symmetry and relax the momentum selection rule. Momentum can be exchanged with the defect, allowing phonons with $\mathbf{q} \neq 0$ to appear in the Raman spectrum.

This mechanism is known as *defect-activated Raman scattering*. It enables first-order Raman features from phonons away from the Brillouin-zone centre and is widely used as a sensitive probe of disorder. The intensity of defect-activated modes typically scales with the defect density.

A classic example is the appearance of Raman features associated with zone-edge phonons, which become allowed when momentum conservation is relaxed. These modes can reveal information about crystal quality, strain fields, or chemically induced disorder.

1.3 Brillouin light scattering

Brillouin light scattering (BLS) is the inelastic scattering of visible light by *acoustic* phonons. Because acoustic phonon frequencies in solids are in the GHz range, the associated frequency shifts of the scattered light are much smaller than in Raman spectroscopy. Nevertheless, the underlying process is analogous: the photon exchanges a quantum of energy and momentum with a lattice vibration, generating Stokes and anti-Stokes sidebands.

Experimental implementation

Modern BLS experiments typically use a single-frequency laser and a high-finesse Fabry–Pérot interferometer to resolve the very small frequency shifts ($\Delta\omega \sim 1\text{--}50 \text{ GHz}$). The laser beam is focused onto the sample, and the scattered light is collected either in a backscattering geometry or at a controlled scattering angle. The interferometer acts as a spectral filter with sub-GHz resolution, enabling detection of the weak Brillouin peaks close to the strong Rayleigh line.

In transparent materials, both surface and bulk acoustic waves can contribute. In opaque materials, only surface or near-surface excitations are typically probed.

Stokes and anti-Stokes Brillouin peaks

The frequency shift of the scattered light is given by

$$\omega_{\text{Stokes}} = \omega_{\text{in}} - \omega_{\mathbf{q}s}, \quad \omega_{\text{anti-Stokes}} = \omega_{\text{in}} + \omega_{\mathbf{q}s},$$

where $\omega_{\mathbf{q}s}$ is the acoustic phonon frequency at the probed wave vector. As in Raman scattering, the anti-Stokes intensity scales with the thermal phonon occupation $n(\omega_{\mathbf{q}s})$, which at GHz frequencies is typically large at room temperature. The Stokes and anti-Stokes Brillouin peaks therefore often have comparable intensities.

Momentum transfer and accessible wave vectors

Because acoustic phonons lie at very small frequencies, Brillouin scattering requires probing phonons at *finite* wave vectors $\mathbf{q}$, not just $\mathbf{q} \approx 0$. The momentum transfer is determined by the scattering geometry:

$$\mathbf{q} = \mathbf{k} - \mathbf{k}', \quad |\mathbf{q}| = \frac{4\pi n}{\lambda} \sin\left(\frac{\theta}{2}\right),$$

where λ is the laser wavelength, n the refractive index, and θ the scattering angle. By varying θ , BLS can measure phonons at different wave vectors along the acoustic branch.

In backscattering ($\theta = 180^\circ$), the magnitude of $\mathbf{q}$ is fixed, while in right-angle (90°) scattering the accessible q is smaller. This geometric control is one of the strengths of BLS.

Acoustic phonons and sound velocity

For small wave vectors, the acoustic phonon dispersion is approximately linear:

$$\omega_{\mathbf{q}s} \approx v_s |\mathbf{q}|,$$

where v_s is the sound velocity of the corresponding mode ($s = \text{LA}$ or TA). Thus the Brillouin shift directly yields the sound velocity:

$$v_s = \frac{\omega_{\mathbf{q}s}}{|\mathbf{q}|}.$$

In anisotropic crystals, different phonon polarizations and propagation directions give access to different elastic constants. BLS therefore provides a non-destructive method to determine v_{LA} and v_{TA} and to extract the elastic tensor.

What BLS measures

Brillouin light scattering provides:

- **frequencies of acoustic phonons** at tunable finite q ,
- **sound velocities** and thus elastic constants,
- **surface acoustic wave** modes in thin films,
- **information on near-surface mechanical properties**.

Together with Raman spectroscopy (optical phonons) and neutron scattering (full dispersion), BLS completes the accessible range of phonon frequencies and wave vectors.

2 Density of states in 1D, 2D and 3D

Many solid state properties — specific heat, thermal conductivity, and electronic transport — are governed by how many excitations are available at a given energy. This is encoded in the *density of states* (DOS) $D(E)$, which is defined such that:

$$dN(E) = D(E) dE = \text{number of states per volume in the energy interval } [E, E + dE]. \quad (1)$$

In crystalline solids it is usually convenient to start in momentum space (more precisely: wave vector $\mathbf{k}$ space) and then transform to energy using the dispersion relation $E(\mathbf{k})$.

2.1 Constant density of states in k -space

We consider a d -dimensional system ($d = 1, 2, 3$) in a box of linear size L (with length L , area $A = L^2$, or volume $V = L^3$ respectively) and impose periodic boundary conditions, such that

$$e^{ik_d(x_d+L)} = e^{ik_dx_d} \quad (2)$$

$$\rightarrow e^{ik_dL} = 1 \quad (3)$$

$$\rightarrow k_dL = 2\pi N \quad (4)$$

Hence, the allowed wave vectors are spaced by

$$dk_d = \frac{2\pi}{L} \quad (5)$$

so that each allowed $\mathbf{k}$ state occupies the same volume in k -space. For electrons we must also include spin degeneracy $g_s = 2$; for phonons we will count the number of phonon branches $g_s \leq 3N_{\text{Basis}}$ separately. The density of states *in k -space* is therefore

$$D(k) = \frac{\text{degeneracy factor}}{\text{real-space volume} \times k\text{-space volume}} = \frac{g_s \cdot L^d}{L^d \cdot (2\pi)^d} = \frac{g_s}{(2\pi)^d}. \quad (6)$$

This result is universal: it depends only on dimension d , not on the dispersion relation. The number of states with wave vector magnitude between k and $k + dk$ is

$$\frac{1}{(\text{real-space volume})} dN = D(k) S_d k^{d-1} dk = \frac{g_s S_d}{(2\pi)^d} k^{d-1} dk, \quad (7)$$

where S_d is the surface of the d -dimensional unit sphere:

$$S_1 = 2, \quad S_2 = 2\pi, \quad S_3 = 4\pi. \quad (8)$$

To obtain the density of states in energy, we change variables from k to E :

$$D(E) dE = \frac{1}{(\text{real-space volume})} dN = \frac{g_s S_d}{(2\pi)^d} k^{d-1} \frac{dk}{dE} dE. \quad (9)$$

All the dependence on the dispersion $E(k)$ is contained in $k(E)$ and dk/dE . Below we apply this formula to (i) a linear dispersion as relevant for acoustic phonons and (ii) a quadratic dispersion as relevant for free electrons. The exponents of E will be crucial for the temperature dependence of specific heat.

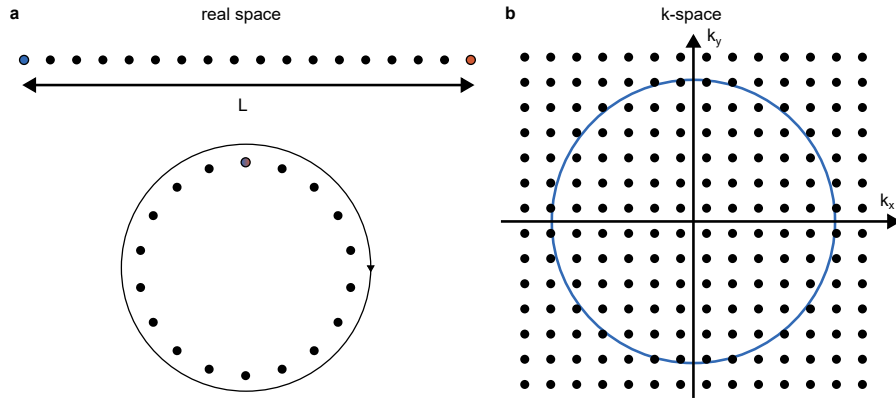


Figure 2: **Counting states.** **a** Periodic boundary conditions in real space. **b** States are distributed equally in k-space.

2.2 Linear dispersion: acoustic phonons

For acoustic phonons near the Brillouin-zone centre we can approximate the dispersion as

$$E = \hbar\omega = \hbar vk, \quad (10)$$

where v is the (sound) velocity of the mode. Inverting this relation gives

$$k(E) = \frac{E}{\hbar v}, \quad \frac{dk}{dE} = \frac{1}{\hbar v}. \quad (11)$$

Inserting into Eq. (9) yields

$$D_{\text{ph}}(E) = \frac{g_s S_d}{(2\pi)^d} \frac{k^{d-1}}{\hbar v} = \frac{g_s S_d}{(2\pi)^d} \frac{1}{(\hbar v)^d} E^{d-1}. \quad (12)$$

Here g_s counts the number of phonon branches (for an isotropic 3D crystal typically $g_s = 3$ acoustic branches).

Explicit forms in 1D, 2D, 3D. Using $S_1 = 2$, $S_2 = 2\pi$ and $S_3 = 4\pi$ we obtain:

- **1D:**

$$D_{\text{ph}}^{(1d)}(E) = \frac{g_s}{\pi} \frac{1}{\hbar v} \propto E^0. \quad (13)$$

The DOS is *constant* for a linear dispersion in 1D.

- **2D:**

$$D_{\text{ph}}^{(2d)}(E) = \frac{g_s}{2\pi} \frac{E}{(\hbar v)^2} \propto E^1. \quad (14)$$

- **3D:**

$$D_{\text{ph}}^{(3d)}(E) = \frac{g_s}{2\pi^2} \frac{E^2}{(\hbar v)^3} \propto E^2. \quad (15)$$

These power laws $g_{\text{ph}}(E) \propto E^{D-1}$ directly lead to the familiar Debye result for the low-temperature specific heat:

$$C_{\text{ph}} \propto T^D \quad (16)$$

in D dimensions (e.g. $C \propto T^3$ in 3D).

2.3 Quadratic dispersion: free electrons

For (nearly) free electrons we use the parabolic dispersion

$$E = \frac{\hbar^2 k^2}{2m^*}, \quad (17)$$

where m^* is the (effective) electron mass. We invert this relation,

$$k(E) = \sqrt{\frac{2m^*E}{\hbar^2}}, \quad \frac{dk}{dE} = \frac{1}{2} \sqrt{\frac{2m^*}{\hbar^2}} E^{-1/2}, \quad (18)$$

and insert it into Eq. (9). This yields

$$\begin{aligned} D_{\text{el}}(E) &= \frac{g_s S_d}{(2\pi)^d} k^{d-1} \frac{dk}{dE} \\ &= \frac{g_s S_d}{(2\pi)^d} \left(\frac{2m^*}{\hbar^2} \right)^{\frac{d-1}{2}} E^{\frac{d-1}{2}} \frac{1}{2} \left(\frac{2m^*}{\hbar^2} \right)^{\frac{1}{2}} E^{-1/2} \\ &= \frac{g_s S_d}{(2\pi)^d} \frac{1}{2} \left(\frac{2m^*}{\hbar^2} \right)^{\frac{d}{2}} E^{\frac{d}{2}-1}. \end{aligned} \quad (19)$$

Explicit forms in 1D, 2D, 3D. Again inserting S_D for $D = 1, 2, 3$:

- **1D:**

$$D_{\text{el}}^{(1\text{d})}(E) = \frac{g_s}{2\pi} \left(\frac{2m^*}{\hbar^2} \right)^{1/2} E^{-1/2} \propto E^{-1/2}. \quad (20)$$

The DOS diverges as $E^{-1/2}$ near the band edge (a 1D Van Hove singularity).

- **2D:**

$$D_{\text{el}}^{(2\text{d})}(E) = \frac{g_s}{4\pi} \left(\frac{2m^*}{\hbar^2} \right) \propto E^0. \quad (21)$$

In 2D, a quadratic band has a *constant* DOS.

- **3D:**

$$D_{\text{el}}^{(3\text{d})}(E) = \frac{g_s}{4\pi^2} \left(\frac{2m^*}{\hbar^2} \right)^{3/2} \sqrt{E} \propto E^{1/2}. \quad (22)$$